

CYBERSECURITY-LABS

INFORMATION SECURITY

LAB REPORT



Práctica: Entorno Ansible

Informe de práctica

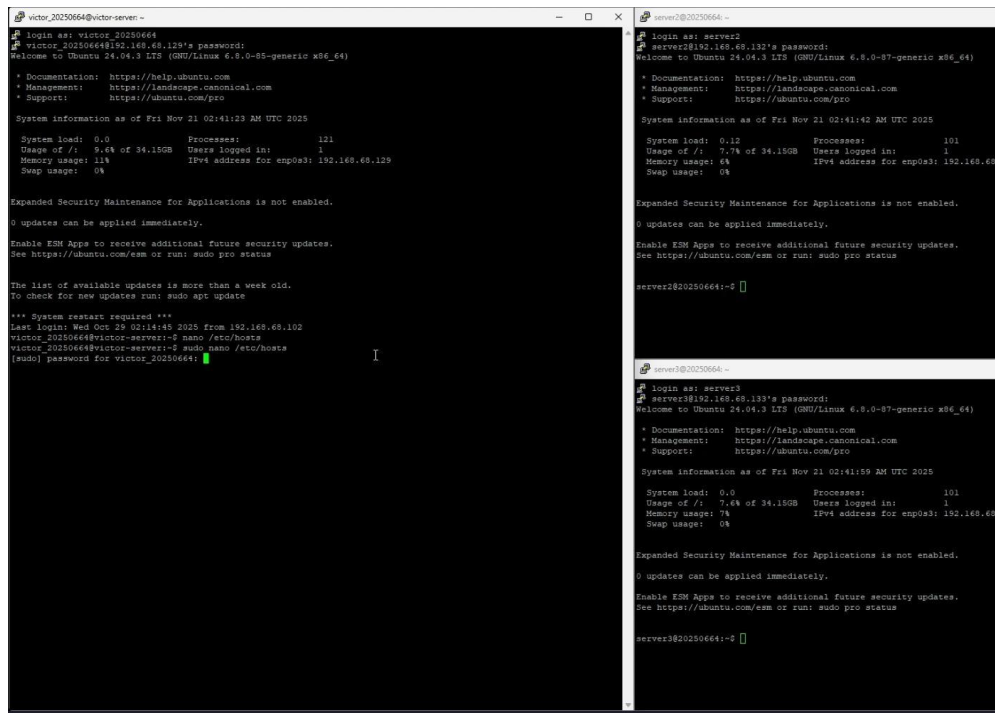
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Information Security

Cybersecurity / Cloud / Networks / GRC

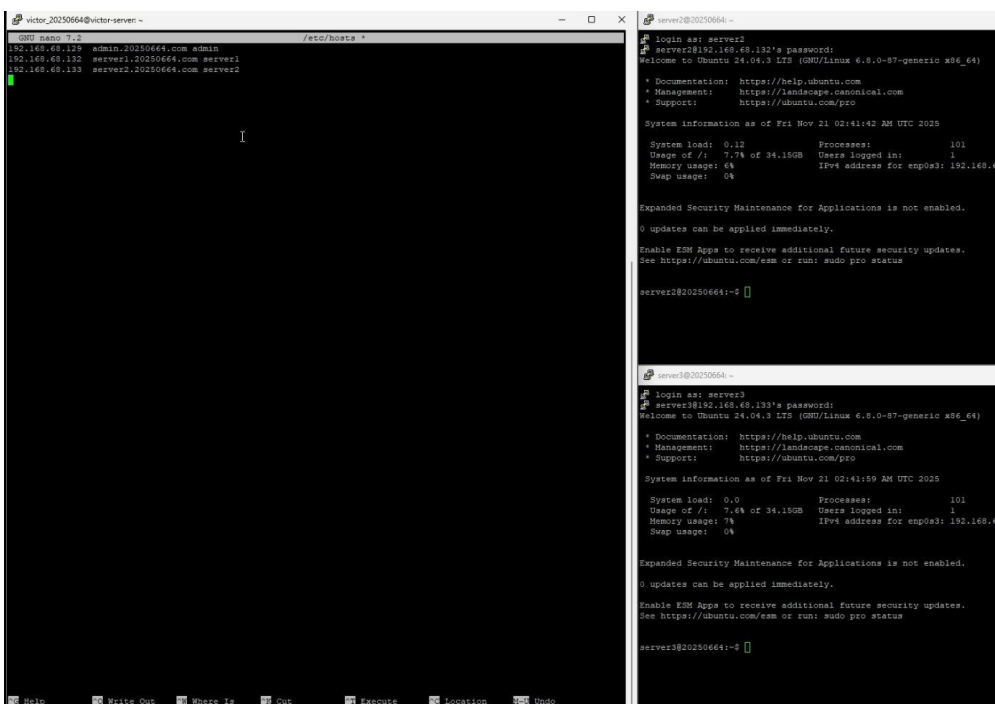


Paso 1: Inicié sesión simultáneamente en mis tres servidores para verificar la conectividad y las direcciones IP asignadas a cada interfaz. En el nodo de control, procedí a ejecutar el comando `sudo nano /etc/hosts` e introduje mi contraseña de sudo; realizo esta acción para editar el archivo de hosts local y asociar las direcciones IP de los nodos remotos con los nombres de dominio `server1.sumatricula.com` y `server2.sumatricula.com`.



The image shows three terminal windows. The left window is the control node (victor_20250664@victor-server) running `sudo nano /etc/hosts`. The middle and right windows are remote servers (server1, server2, and server3) showing login prompts and system information. The system information for all servers is as of Fri Nov 21 02:41:23 AM UTC 2025. The control node shows system load: 0.0, Processes: 121, Users logged in: 1, Memory usage: 11%, Swap usage: 0%, and IP address for enp0s3: 192.168.68.129. The remote servers show system load: 0.12, Processes: 101, Users logged in: 1, Memory usage: 6%, Swap usage: 0%, and IP address for enp0s3: 192.168.68.132 (server1), 192.168.68.133 (server2), and 192.168.68.134 (server3).

Paso 2: Dentro del editor de texto, agregué las líneas necesarias relacionando las direcciones IP privadas con los nombres de dominios requeridos: `Admin.sumatricula.com`, `server1.sumatricula.com` y `server2.sumatricula.com`. Una vez ingresados los datos, procedí a guardar el archivo confirmando la escritura del buffer con `Y` y salí del editor para aplicar la configuración de resolución de nombres local en el servidor de control.



The image shows three terminal windows. The left window is the control node (victor_20250664@victor-server) running `sudo nano /etc/hosts`. The middle and right windows are remote servers (server1, server2, and server3) showing login prompts and system information. The system information for all servers is as of Fri Nov 21 02:41:23 AM UTC 2025. The control node shows system load: 0.0, Processes: 121, Users logged in: 1, Memory usage: 11%, Swap usage: 0%, and IP address for enp0s3: 192.168.68.129. The remote servers show system load: 0.12, Processes: 101, Users logged in: 1, Memory usage: 6%, Swap usage: 0%, and IP address for enp0s3: 192.168.68.132 (server1), 192.168.68.133 (server2), and 192.168.68.134 (server3).

Paso 3: Verifiqué la correcta resolución de nombres desde el nodo de control lanzando un ping -c 3 hacia server1 y server2, confirmando que responden adecuadamente. Simultáneamente, en los hosts administrados, ejecuté el comando sudo nano /etc/hosts para editar sus archivos de configuración local, asegurando así que la resolución de nombres sea consistente en todo el entorno del cluster.

```
victor.2025064@victor-seven:~$ sudo useradd -s /bin/bash -m -s s-admin useradd: invalid shell '/bin/bash'
victor.2025064@victor-seven:~$ sudo useradd -m -s s-admin useradd: password is required
victor.2025064@victor-seven:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
victor.2025064@victor-seven:~$ cat /etc/passwd
victor.2025064@victor-seven:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
victor.2025064@victor-seven:~$ sudo nano /etc/hosts
1
victor.2025064@victor-seven:~$
```

```
server2.2025064@server2:~$ ip a
1: lo: <LOOPBACK>,UP,LOWER_UP,MTU 65536 qdisc noqueue state UNKNOWN scope global
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefix
        valid_lft forever preferred_lft forever
2: ens3: <VMXNET3>,MULTICAST,UP,LOWER_UP,MTU 1500 qdisc fq_codel state UP
    link/ether 08:00:27:fc8b:86c7:74 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global
        valid_lft 4754sec preferred_lft 4754sec
    inet6 fe80::a00:127:fc8b:86c7:74 scope link
        valid_lft forever preferred_lft forever
server2.2025064@server2:~$ sudo useradd -m s-admin
server2.2025064@server2:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
server2.2025064@server2:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
server2.2025064@server2:~$
```

```
server2.2025064@server2:~$ ip a
1: lo: <LOOPBACK>,UP,LOWER_UP,MTU 65536 qdisc noqueue state UNKNOWN scope global
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefix
        valid_lft forever preferred_lft forever
2: ens3: <VMXNET3>,MULTICAST,UP,LOWER_UP,MTU 1500 qdisc fq_codel state UP
    link/ether 08:00:27:fc8b:86c7:74 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global
        valid_lft 4754sec preferred_lft 4754sec
    inet6 fe80::a00:127:fc8b:86c7:74 scope link
        valid_lft forever preferred_lft forever
server2.2025064@server2:~$ sudo useradd -m s-admin
server2.2025064@server2:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
server2.2025064@server2:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
server2.2025064@server2:~$
```

Paso 4: En todos los nodos del clúster, creé el usuario administrativo con el comando sudo useradd -m s-admin y definí su credencial de acceso mediante sudo passwd s-admin. Acto seguido, habilité los permisos de superusuario sin solicitud de contraseña para este usuario ejecutando la instrucción echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin, garantizando así la automatización futura con Ansible.

```
victor.2025064@victor-seven:~$ login as: victor.2025064
victor.2025064@victor-seven:~$ sudo useradd -m -s s-admin useradd: password is required
victor.2025064@victor-seven:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
victor.2025064@victor-seven:~$ cat /etc/passwd
victor.2025064@victor-seven:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
victor.2025064@victor-seven:~$
```

```
server2.2025064@server2:~$ ip a
1: lo: <LOOPBACK>,UP,LOWER_UP,MTU 65536 qdisc noqueue state UNKNOWN scope global
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefix
        valid_lft forever preferred_lft forever
2: ens3: <VMXNET3>,MULTICAST,UP,LOWER_UP,MTU 1500 qdisc fq_codel state UP
    link/ether 08:00:27:fc8b:86c7:74 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global
        valid_lft 4754sec preferred_lft 4754sec
    inet6 fe80::a00:127:fc8b:86c7:74 scope link
        valid_lft forever preferred_lft forever
server2.2025064@server2:~$ sudo useradd -m s-admin
server2.2025064@server2:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
server2.2025064@server2:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
server2.2025064@server2:~$
```

```
server2.2025064@server2:~$ ip a
1: lo: <LOOPBACK>,UP,LOWER_UP,MTU 65536 qdisc noqueue state UNKNOWN scope global
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefix
        valid_lft forever preferred_lft forever
2: ens3: <VMXNET3>,MULTICAST,UP,LOWER_UP,MTU 1500 qdisc fq_codel state UP
    link/ether 08:00:27:fc8b:86c7:74 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global
        valid_lft 4754sec preferred_lft 4754sec
    inet6 fe80::a00:127:fc8b:86c7:74 scope link
        valid_lft forever preferred_lft forever
server2.2025064@server2:~$ sudo useradd -m s-admin
server2.2025064@server2:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
server2.2025064@server2:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
server2.2025064@server2:~$
```


Paso 7: Actualicé los repositorios del sistema y procedí a instalar el motor de automatización en el nodo de control ejecutando el comando `sudo apt install -y ansible`. Una vez finalizada la instalación, verifiqué la versión correcta con `ansible --version` y preparé mi espacio de trabajo creando el directorio del proyecto mediante `mkdir -p ~/ansible_project`, accediendo inmediatamente a él con `cd ~/ansible_project`.

```
victor_20250664@victor-server:~$ sudo apt update
sudo apt install -y ansible
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://do.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://do.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [11,311 kB]
Hit:5 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [216 kB]
Hit:6 http://do.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Hit:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Hit:8 http://security.ubuntu.com/ubuntu noble-updates/main Translation-en [5,452 B]
Hit:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,146 kB]
Hit:10 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,412 kB]
Hit:11 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [488 kB]
Hit:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Hit:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [909 kB]
Hit:14 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [305 kB]
Hit:15 http://do.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [302 kB]
Hit:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Hit:17 http://security.ubuntu.com/ubuntu noble-security/universe amd64 C-n-f Metadata [15.4 kB]
Hit:18 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5,954 B]
Hit:19 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Hit:20 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Hit:21 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 C-n-f Metadata [15.7 kB]
Hit:22 http://do.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,389 kB]
Hit:23 http://do.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [519 kB]
Hit:24 http://do.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [213 B]
Hit:25 http://do.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,300 kB]
Hit:26 http://do.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [304 kB]
Hit:27 http://do.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Hit:28 http://do.archive.ubuntu.com/ubuntu noble-updates/universe amd64 C-n-f Metadata [14.4 kB]
Hit:29 http://do.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [5,808 B]
Hit:30 http://do.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Hit:31 http://do.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,124 B]
Hit:32 http://do.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Hit:33 http://do.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Hit:34 http://do.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 12.9 MB in 3s (4,017 KB/s)

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfile
  python3-nlme-auth python3-passlib python3-requests-nlme python3-resolvlib python3-selinux python3-singlejson
  python3-winnm python3-winnm
Suggested packages:
  cowsay empsay python3-trio python3-sicp python3-h2 python3-httpx python3-httpcore python-lockfile-doc
The following NEW packages will be installed:
  ansible ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfile
  python3-nlme-auth python3-passlib python3-requests-nlme python3-resolvlib python3-selinux python3-singlejson
  python3-winnm python3-winnm
0 upgraded, 15 newly installed, 0 to remove and 20 not upgraded.
Need to get 19.5 MB of archives.
After this operation, 315 MB of additional disk space will be used.
Get:1 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolvlib 1.0.0-1 [25.7 kB]
Get:2 http://do.archive.ubuntu.com/ubuntu noble/main amd64 python3-dnspython 2.6.1-1ubuntu1 [163 kB]
Get:3 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 ansible-core 2.16.3-0ubuntu2 [1,280 kB]
Get:4 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 ansible 2.16.3-0ubuntu2 [16.4 MB]
debconf: 15.0 KB/16.4 MB OK

Unpacking python3-selinux (3.5-2ubuntu2.1) ...
Selecting previously unselected package python3-winnm.
Preparing to unpack .../12-python3-winnm_0.13.0-0ubuntu0.24.04.1_all.deb ...
Unpacking python3-winnm (0.13.0-0ubuntu0.24.04.1) ...
Selecting previously unselected package python3-winnm.
Preparing to unpack .../13-python3-winnm_0.13.0-2_all.deb ...
Unpacking python3-winnm (0.13.0-2) ...
Selecting previously unselected package python3-passlib.
Preparing to unpack .../14-python3-passlib_1.7.4-4_all.deb ...
Unpacking python3-passlib (1.7.4-4) ...
Setting up python3-lockfile (1.0.12-2.3) ...
Setting up python3-passlib (1.7.4-4) ...
Setting up python3-nlme-auth (0.13.0-0ubuntu0.24.04.1) ...
Setting up python3-resolvlib (1.0.0-1) ...
Setting up python3-kerberos (3.15.2-1ubuntu1) ...
Setting up python3-singlejson (0.13.0-0ubuntu0.24.04.1) ...
Setting up python3-dnspython (2.6.1-1ubuntu1) ...
Setting up python3-selinux (3.5-2ubuntu2.1) ...
Setting up python3-argcomplete (2.1.4-0ubuntu0.24.04.1) ...
Setting up python3-requests-nlme (1.1.0-3) ...
Setting up ansible-core (2.16.3-0ubuntu2) ...
Setting up python3-libcloud (3.4.1-9) ...
Setting up python3-winnm (0.13.0-2) ...
Setting up ansible (2.16.3-0ubuntu2) ...
Processing triggers for man-db (2.12.0-4build1) ...
Scanning processes...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-36-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-36-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
  /etc/needrestart/restart.d/dbus.service
  systemd-logind.service
  systemctl restart systemd-updates.service

No containers need to be restarted.

User sessions running outdated binaries:
  victor_20250664 @ session #1: login[943]
  victor_20250664 @ user manager service: systemd[1160]

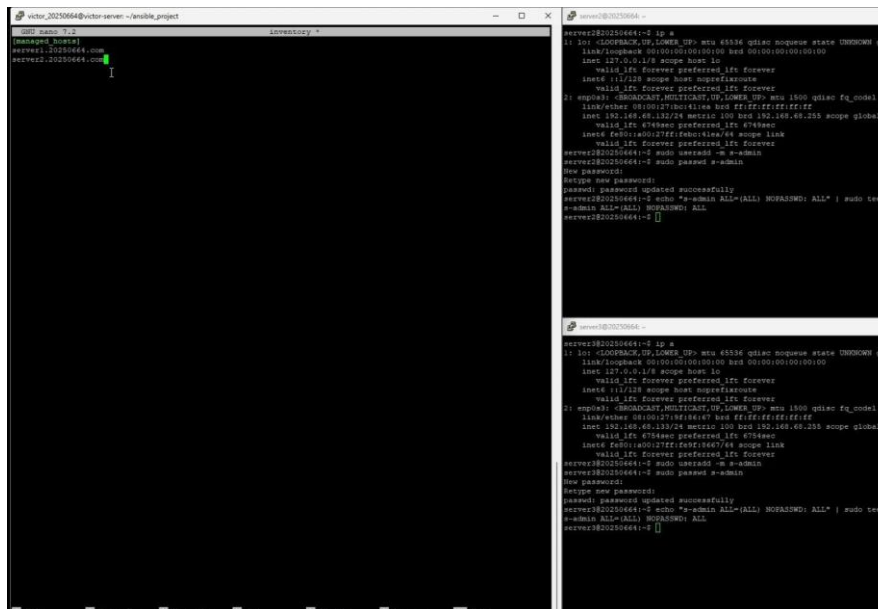
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ansible (core 2.16.3)
  config file = None
  configured module search path = ['/home/victor_20250664/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/victor_20250664/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Aug 14 2025, 17:47:21) [GCC 13.3.0] (/usr/bin/python3)
  libyaml = True
victor_20250664@victor-server:~$ mkdir -p ~/ansible_project
victor_20250664@victor-server:~$ cd ~/ansible_project/

server20250664:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
    link/ether 08:00:27:9f:86:67 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global
        valid_lft 6754sec preferred_lft 6754sec
    inet6 fe80::a00:27ff:fe9f:8667/64 scope link
        valid_lft forever preferred_lft forever
server20250664:~$ sudo useradd -m s-admin
server20250664:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
server20250664:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee
s-admin ALL=(ALL) NOPASSWD: ALL
server20250664:~$

server30250664:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
    link/ether 08:00:27:9f:86:67 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global
        valid_lft 6754sec preferred_lft 6754sec
    inet6 fe80::a00:27ff:fe9f:8667/64 scope link
        valid_lft forever preferred_lft forever
server30250664:~$ sudo useradd -m s-admin
server30250664:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
server30250664:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee
s-admin ALL=(ALL) NOPASSWD: ALL
server30250664:~$

server3820250664:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
    link/ether 08:00:27:9f:86:67 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global
        valid_lft 6754sec preferred_lft 6754sec
    inet6 fe80::a00:27ff:fe9f:8667/64 scope link
        valid_lft forever preferred_lft forever
server3820250664:~$ sudo useradd -m s-admin
server3820250664:~$ sudo passwd s-admin
New password:
passwd: password updated successfully
server3820250664:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee
s-admin ALL=(ALL) NOPASSWD: ALL
server3820250664:~$
```

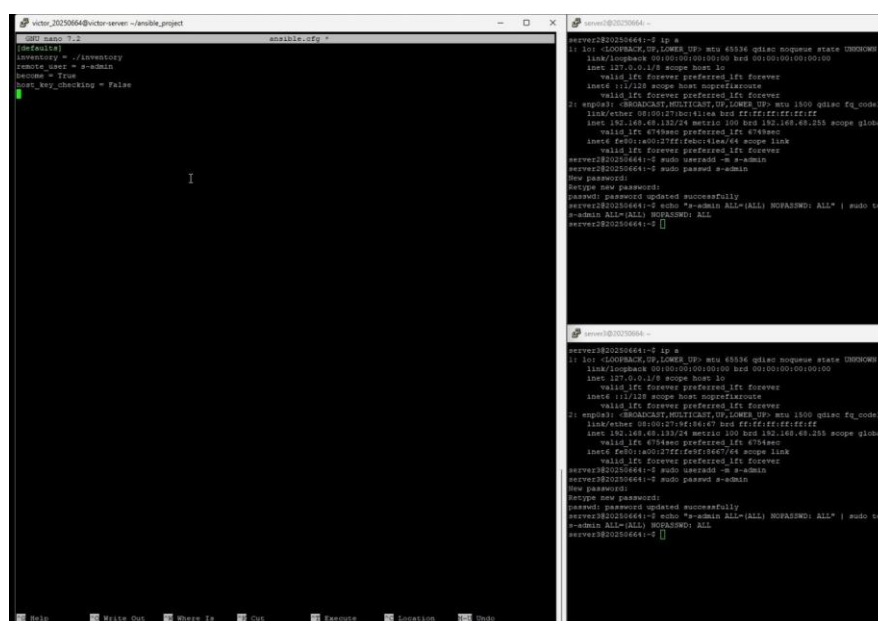
Paso 8: Procedí a crear el archivo de inventario dentro de mi directorio de proyecto ejecutando el comando `nano inventory`. En su interior, definí el grupo `[managed_hosts]` y listé debajo los nombres de dominio `server1.sumatricula.com` y `server2.sumatricula.com`, estableciendo así los objetivos que serán gestionados centralizadamente por Ansible.



```
[victor_2025064@victor-server: ~/ansible_project]$ nano inventory
[managed_hosts]
server1.2025064.com
server2.2025064.com

[victor_2025064@victor-server: ~/ansible_project]$ cat inventory
[managed_hosts]
server1.2025064.com
server2.2025064.com
```

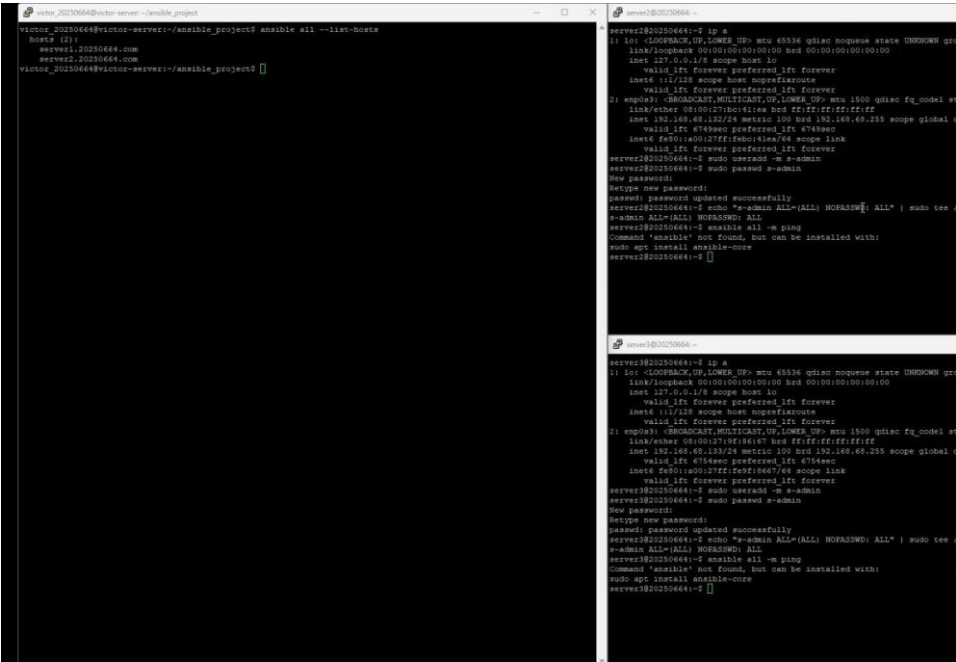
Paso 9: Creé el archivo de configuración principal ejecutando `nano ansible.cfg` para definir los parámetros globales del proyecto. Dentro de la sección `[defaults]`, especifiqué la ruta de mi inventario, configuré a `s-admin` como el usuario remoto predeterminado y activé la escalada de privilegios automática con `become = True`, añadiendo también `host_key_checking = False` para agilizar la conexión sin validaciones manuales de claves.



```
[victor_2025064@victor-server: ~/ansible_project]$ nano ansible.cfg
[defaults]
inventory = ./inventory
remote_user = s-admin
become = True
host_key_checking = False

[victor_2025064@victor-server: ~/ansible_project]$ cat ansible.cfg
[defaults]
inventory = ./inventory
remote_user = s-admin
become = True
host_key_checking = False
```

Paso 10: Validé la correcta integración del inventario ejecutando el comando ansible all --list-hosts en el directorio del proyecto, lo que me permitió visualizar los hosts disponibles. La salida en terminal confirmó la presencia de server1.sumatricula.com y server2.sumatricula.com, asegurando que mi entorno Ansible reconoce perfectamente los nodos administrados que configuré previamente en el archivo inventory.

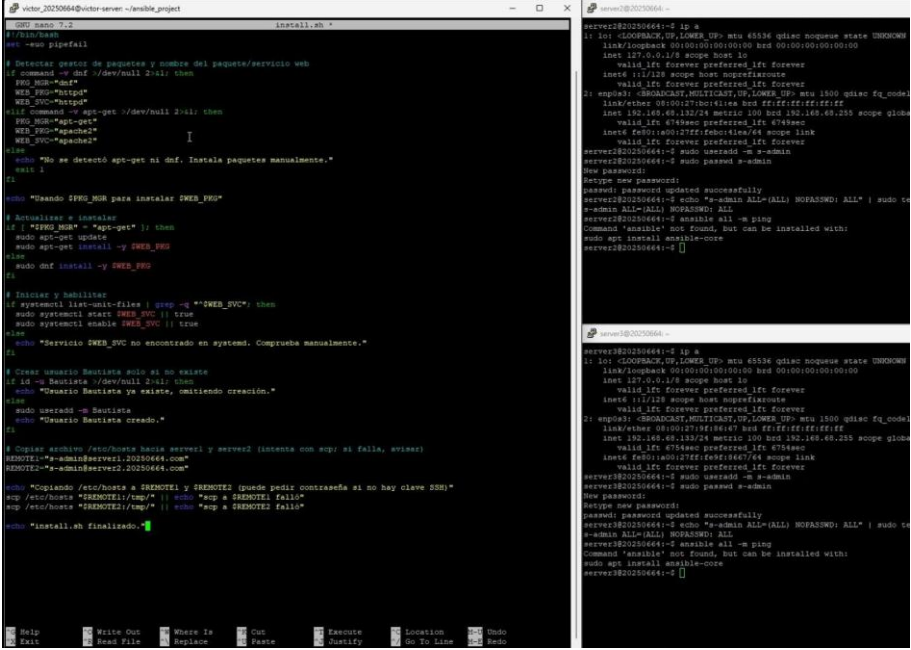


```
victor_20250664@victor-server: ~/ansible_project
victor_20250664@victor-server:~/ansible_project$ ansible all --list-hosts
hosts (2):
  server1.20250664.com
  server2.20250664.com
victor_20250664@victor-server:~/ansible_project$

server2[20250664]:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qlen 1000 state UNKNOWN scope global noprefixroute
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qlen 1000 state UP scope global dynamic noprefixroute
    link/ether 08:00:27:1b:c4:1e brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global dynamic noprefixroute
        valid_lft 4754sec preferred_lft 4754sec
    inet6 fe80::a00:127ff:fe01:1e67/64 scope link
        valid_lft forever preferred_lft forever
server2[20250664]:~$ sudo useradd -m s-admin
server2[20250664]:~$ sudo passwd s-admin
New password:
Retype new password:
passwd: password updated successfully
server2[20250664]:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
s-admin ALL=(ALL) NOPASSWD: ALL
server2[20250664]:~$ ansible all -m ping
Command 'ansible' not found, but can be installed with:
sudo apt install ansible-core
server2[20250664]:~$

server1[20250664]:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qlen 1000 state UNKNOWN scope global noprefixroute
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qlen 1000 state UP scope global dynamic noprefixroute
    link/ether 08:00:27:1b:c4:1e brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global dynamic noprefixroute
        valid_lft 4754sec preferred_lft 4754sec
    inet6 fe80::a00:127ff:fe01:1e67/64 scope link
        valid_lft forever preferred_lft forever
server1[20250664]:~$ sudo useradd -m s-admin
server1[20250664]:~$ sudo passwd s-admin
New password:
Retype new password:
passwd: password updated successfully
server1[20250664]:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
s-admin ALL=(ALL) NOPASSWD: ALL
server1[20250664]:~$ ansible all -m ping
Command 'ansible' not found, but can be installed with:
sudo apt install ansible-core
server1[20250664]:~$
```

Paso 11: Procedí a crear el script de automatización ejecutando nano install.sh en el nodo de control. En este fichero, redacté la lógica bash necesaria para instalar y habilitar el servidor web, crear el usuario **Bautista** y transferir el archivo /etc/hosts al directorio /tmp/ de los nodos remotos mediante scp, guardando posteriormente los cambios para asegurar que los hosts administrados cumplan con los requisitos.



```
victor_20250664@victor-server:~/ansible_project
# nano install.sh
#!/bin/bash
set -e

# Detectar directorio de paquetes y nombre del paquete/servicio web
PKG_MGR="apt"
WEB_PKG="httpd"
WEB_USER="httpd"

if command -v apt-get > /dev/null 2>&1; then
    PKG_MGR="apt-get"
    WEB_PKG="apache2"
    WEB_USER="apache2"
else
    PKG_MGR="yum"
    WEB_PKG="httpd"
    WEB_USER="httpd"
fi

# Instalar y habilitar
systemctl list-unit-files | grep -q "${WEB_PKG}.service"
systemctl start ${WEB_PKG}.service
systemctl enable ${WEB_PKG}.service

# Crear usuario Bautista si no existe
if id -u Bautista > /dev/null 2>&1; then
    echo "Usuario Bautista ya existe, omitiendo creación."
else
    useradd -m -s /bin/bash Bautista
    echo "Usuario Bautista creado."
fi

# Copiar archivo /etc/hosts hacia server1 y server2 (intenta con scp; si falla, mv)
REMOTE1="s-admin@server1.20250664.com"
REMOTE2="s-admin@server2.20250664.com"

# Copiando /etc/hosts hacia REMOTE1 y REMOTE2 (puede pedir contraseña si no hay clave SSH)
scp /etc/hosts "${REMOTE1}:/tmp/" || echo "scp a REMOTE1 fallo"
scp /etc/hosts "${REMOTE2}:/tmp/" || echo "scp a REMOTE2 fallo"

# Finalizando.

server2[20250664]:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qlen 1000 state UNKNOWN scope global noprefixroute
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qlen 1000 state UP scope global dynamic noprefixroute
    link/ether 08:00:27:1b:c4:1e brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.132/24 metric 100 brd 192.168.68.255 scope global dynamic noprefixroute
        valid_lft 4754sec preferred_lft 4754sec
    inet6 fe80::a00:127ff:fe01:1e67/64 scope link
        valid_lft forever preferred_lft forever
server2[20250664]:~$ sudo useradd -m s-admin
server2[20250664]:~$ sudo passwd s-admin
New password:
Retype new password:
passwd: password updated successfully
server2[20250664]:~$ echo "s-admin ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/s-admin
s-admin ALL=(ALL) NOPASSWD: ALL
server2[20250664]:~$ ansible all -m ping
Command 'ansible' not found, but can be installed with:
sudo apt install ansible-core
server2[20250664]:~$
```

Paso 12: Habilité los permisos de ejecución en el archivo recién creado utilizando el comando `chmod +x install.sh` y procedí a iniciar el despliegue con `./install.sh`. Inmediatamente, el script detectó el gestor de paquetes del sistema y comenzó la descarga e instalación automatizada del servidor web `apache2`, mostrando en la terminal el desempaquetado y configuración de las librerías necesarias

```
Victor_20250664@Victor-server: ~/ansible/project
root@127:
server1-20250664.com
server2-20250664.com
Victor_20250664@Victor-server:~/ansible/project$ nano install.sh
Victor_20250664@Victor-server:~/ansible/project$ chmod +x install.sh
Victor_20250664@Victor-server:~/ansible/project$ ./install.sh
[...]
```

Paso 13: Observé cómo el script finalizaba habilitando los módulos de Apache y confirmaba la creación exitosa del usuario "Bautista". Por último, el sistema procedió a distribuir el archivo de configuración de hosts a los servidores remotos `scp`; en este punto, ingresé la contraseña del usuario `s-admin` cuando fue solicitada para autorizar la escritura segura en el directorio temporal de `server1` y finalizar la práctica.

```
Victor_20250664@Victor-server: ~/ansible/project
[...]
```